

Lakshima Madurangi

Software Engineering Graduate | AI, Web Development
& Data Analytics

076-0479712 | mhlakshima@gmail.com

 LinkedIn: [linkedin.com/in/m-h-lakshima-3a590728](https://www.linkedin.com/in/m-h-lakshima-3a590728)

 GitHub: github.com/Lakshima409

PORTFOLIO



PROFILE

Software Engineering graduate with a BEng (Hons) in Software Engineering, an HND in Computing, and hands-on experience developing AI-powered, full-stack and desktop applications through academic projects. Skilled in Python, C#, JavaScript, HTML, CSS, React.js, Node.js, PHP, SQL and data analytics, with experience in machine learning concepts, REST APIs, database systems, software testing and SEO-focused web development. Curious and quick to learn new technologies, with a strong interest in AI tools, automation and modern digital workflows. Comfortable working independently or collaboratively, researching unfamiliar tools, checking technical output carefully and continuously improving solutions.

EDUCATION AND TRAINING

Bachelor of Engineering (Hons) in Software Engineering – Upper Second Class Honours **2025-2026**

London Metropolitan University

Relevant Modules:

- Advanced Software Engineering
- Application Development
- Artificial Intelligence
- Project Analysis and Practice
- Final Year Research Project

Higher National Diploma (HND) in Computing
ESOFT Metro Campus

2023 - 2024
September

Level 3 Diploma in Information
Technology (DITEC)
ESOFT Metro Campus

2023

English Certificate Course
NIBM

2022

Sangamiththa College
O/L - A7 B1 C1
A/L - 2S F(Maths Stream)

2007 - 2023

Currently Studying

- Google Data Analytics Professional Certificate (Coursera)
- Google Advanced Data Analytics Certificate (Coursera)
- Cisco Certification Courses

2026

TECHNICAL SKILLS

- AI & Data:

Machine Learning Concepts, AI Applications, Data Analytics, SPSS, Data Visualization, Predictive Analytics

- Programming:

Python, C#, Java, JavaScript, PHP, SQL

- Web Development:

HTML, CSS, React.js, Node.js, Express.js, Flask, ASP.NET MVC, Bootstrap 5

- APIs & Development:

REST APIs, Full-Stack Development, Authentication, Role-Based Access Control

- Databases:

MySQL, SQL Server, Database Design, Relational Databases

- Software Engineering:

System Analysis, Requirements Engineering, Software Testing, Agile, UML, OOP, Design Patterns

- Digital & Other:

SEO, Microsoft Office, Research, Documentation, Project Management

- AI & Productivity Tools:

ChatGPT, Claude, Gemini, Blackbox AI, GitHub Copilot, Cursor, AI-assisted research, AI-assisted development, technical problem solving, documentation and content drafting

SOFT SKILLS

- Quick learner
- Problem solving
- Attention to detail
- Independent work
- Team collaboration
- Written communication
- Time management
- Continuous learning

LANGUAGES

- Sinhala (native)
- English

ACADEMIC PROJECTS

PantryPal – AI Recipe Recommendation & Meal Planning System | React.js, Node.js, Express.js, Python, Flask, MySQL, Machine Learning, REST APIs

Final-Year Research Project Assessment: Proposal — B | Interim Report — B | Final Report — A

- Developed a full-stack AI-powered web application providing personalized recipe recommendations based on users' pantry items, dietary preferences, and previous interactions.
- Designed the system architecture, database structure, application workflow, and frontend/backend components using React.js, Node.js, Express.js, Flask, and MySQL.
- Implemented a hybrid recommendation approach combining content-based filtering, collaborative filtering, TF-IDF, cosine similarity, and machine-learning techniques to generate personalized recipe recommendations.
- Developed features including pantry management, dietary personalization, meal planning, user authentication, and automated shopping-list generation through REST API integration.
- Conducted software testing, system evaluation, and research analysis to assess functionality, recommendation performance, and overall system effectiveness.

eShift – Household Goods Transport Management System | C#, Windows Forms, SQL Server

- Developed a desktop transport management system to automate booking, vehicle and driver management, route planning, delivery tracking, and reporting.
- Implemented role-based authentication, secure password handling, database normalization, and layered architecture.
- Applied object-oriented programming, design patterns, unit testing, and business logic implementation to develop a maintainable application.

Luxora Hotel Suite – Property Management System | PHP, MySQL, Bootstrap 5, REST APIs

- Collaborated in a four-person Agile team to develop a full-stack hotel reservation and property management system.
- Developed billing, check-in/check-out, dynamic invoicing, split payments, tax calculations, late-checkout penalties, and reservation handling.
- Implemented role-based access control, audit logging, database normalization, and responsive web interfaces.

Event Ticket Online Web Application | ASP.NET MVC, C#, SQL, Full-Stack Development

- Developed a scalable event-ticketing platform supporting event management, venue configuration, ticket booking, seat selection, QR-code e-tickets, and customer booking history.
- Implemented role-based authentication, payment processing, notifications, loyalty features, promotional codes, and analytics functionality.
- Applied clean architecture and modular design principles to support maintainability and scalability.

ACADEMIC PROJECTS

AI Mental Health Chatbot | NLP, AI, Knowledge-Based Systems

- Collaborated on a conversational AI chatbot using natural language processing, knowledge-based reasoning, and context-aware response generation.
- Contributed to system analysis, architecture, implementation, knowledge-base design, testing, and technical documentation.
- Implemented conversation tracking and persistent user-data functionality as part of a team-based university project.

Big Data in Retail – Research Project | SPSS, Python, Data Analytics

- Investigated the impact of big data on retail marketing strategies, consumer behaviour, segmentation, and personalization.
- Analyzed quantitative research data using SPSS, including descriptive statistics, correlation, regression, and Cronbach's alpha reliability analysis.
- Evaluated predictive analytics and ethical considerations related to consumer data, privacy, and responsible data usage in retail.

Sampath Food City Management System | C#, OOP, Design Patterns, UML

- Developed a scalable retail management system using object-oriented programming and software engineering principles.
- Applied SOLID principles and Factory and Strategy design patterns to create maintainable, flexible, and reusable system components.
- Designed UML models and applied software architecture principles to support modularity, scalability, and maintainability.

Bus Seat Reservation System – Data Structures & Algorithms | Java, Data Structures, Algorithms

- Developed a Java-based bus seat reservation system for managing passenger bookings and seat allocation.
- Implemented stacks, queues, linked lists, and graphs to efficiently manage dynamic reservation and passenger data.
- Applied graph-based shortest-path algorithms and algorithmic problem-solving techniques to support efficient route management.

Photographer's Website - HTML, CSS, JavaScript, PHP, MySQL, SEO

- Designed and developed a responsive photographer portfolio website using HTML, CSS, JavaScript, PHP and MySQL, with a focus on usability and responsive presentation.
- Implemented CMS-based content management and structured the website to support efficient content updates and maintenance.
- Applied basic SEO optimization techniques to improve website structure, content visibility and search-engine accessibility.